

CL 205: Artificial Intelligence & Data Science

Course Project

Department of Chemical Engineering

September 26, 2025

Overview

This course project is an important component of the course, wherein you can try out the various concepts and techniques learned in the classroom and beyond. You are free to choose the dataset from any source and you will use it as a testbed. However, you must understand enough of the underlying story so that you can interpret your results. Some sources for data are

- Kaggle
- Hugging face
- AIKosh
- Espncriinfo
- ...

1 Work Elements

Your project work should include the following three elements:

1. **Visualization of dataset:** Visualize your data using tools such as box plot, histogram, pie chart, scatter plot, bar chart, heat map, donut chart, etc. List down salient observations from the visualization.
2. **Descriptive statistics:** Summarize the data using descriptors such as sample mean, median variance, skew, correlations, range, etc.

3. **Distribution of data:** Q-Q plots is a great graphical way of assessing if the data belong to a particular distribution (such as exponential, Gaussian, binomial, etc). Take few key variables from your dataset and assess the underlying distribution using Q-Q plots for different distributions.
4. **Inferential statistics:**
 - (a) **Estimation:** For a parameter of interest connected with the story, obtain its point estimate and interval estimate
 - (b) **Hypothesis test:** Set up a hypothesis test for a certain assertion on the parameter value and test it at an assumed significance level
 - (c) **Regression:** Create a model to predict a variable of interest (dependent variable) using other variables (independent variables)
 - (d) **Classification:** Create a model to assign a suitable class label to the model input (if class is not part of the dataset, define the class suitably)

2 Tasks

Please do the following:

1. Make a team of four members each and name your team
2. Identify a dataset that your team will work on and the analysis that you will perform
3. Submit a one-page report on Moodle with the following (**Deadline: October 12, 11:59pm**):
 - Details of all team members
 - Brief description of dataset
 - List of analysis that you will carry out with the dataset
4. Upload (1) final project report and (2) codes used on Moodle with the following sections (**Deadline: November 07, 11:59pm**):
 - Introduction to dataset and the story
 - Visual analytics and resulting observations about the story
 - Descriptive statistics and conclusions about the story
 - Inferential statistics with what inferences one can draw about the story